

Name of the Student	N.AKSHAYA
Reg. No	192425129
GitHub Link	https://github.com/akshaya24032007/MLA0405-DeepLearning/tree/main/Assessment/CO4

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

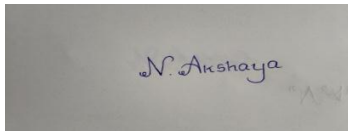
Course Outcome Weightage: 10%
Assessment Tool Weightage: 40%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I **N.AKSHAYA(192425129)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student:N.AKSHAYA

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Q1. Transfer Learning

A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

Answer:

A pre-trained **ResNet50** CNN can be used for skin cancer classification because it has strong feature extraction capability and can learn useful visual features such as edges, textures, shapes, and lesion patterns from limited medical data.

Proposed Design:

- Input: Dermoscopic image resized to $224 \times 224 \times 3$.
- Use **ImageNet-pretrained ResNet50** as the feature extractor.
- Freeze the initial convolutional layers because they learn general features.
- Freeze most intermediate layers initially.
- Replace the original classification layer with:
 - Global Average Pooling
 - Dense layer with ReLU
 - Dropout for regularization
 - Output layer with Softmax/Sigmoid depending on the number of classes.
- Train only the new classification layers initially.
- Subsequently unfreeze the final few ResNet50 blocks and fine-tune them using a **small learning rate**.
- Apply augmentation such as rotation, flipping, zooming, and brightness variation.
- Evaluate using **accuracy, precision, recall, F1-score, sensitivity, specificity, and confusion matrix**.

Architecture:

Dermoscopic Image $\rightarrow$ ResNet50 $\rightarrow$ Feature Extraction $\rightarrow$ Global Average Pooling $\rightarrow$ Dense + ReLU $\rightarrow$ Dropout $\rightarrow$ Softmax $\rightarrow$ Skin Cancer Class

Justification:

Transfer learning reduces training time and the requirement for large labeled datasets. Freezing early layers preserves general visual features, while fine-tuning deeper layers adapts the model to skin-lesion characteristics. Data augmentation and dropout reduce overfitting. This directly addresses the limited labeled-data constraint specified in the question.

Q2. DenseNet and PixelNet

A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational efficiency.

Answer:

A combined **DenseNet + PixelNet** architecture can be designed for pixel-level road-scene segmentation. DenseNet is used for powerful feature extraction and feature reuse, while PixelNet performs pixel-level prediction.

Proposed Architecture:

Road Image → DenseNet Backbone → Multi-Level Feature Extraction → Feature Fusion → PixelNet → Pixel-wise Classification → Segmentation Map

Working:

1. Input road image is resized and normalized.
2. **DenseNet** extracts hierarchical features.
3. Dense connections allow each layer to receive features from previous layers.
4. Important low-level and high-level features are fused.
5. The fused representation is passed to the **PixelNet segmentation module**.
6. PixelNet predicts a class for every pixel.
7. The final segmentation map identifies:
 - Road
 - Pedestrians
 - Vehicles
 - Traffic signs
8. Post-processing can be applied to remove small segmentation errors.

Advantages:

- Dense connections improve **feature reuse**.
- Reusing features reduces redundant computation and parameters.
- Multi-level features improve boundary detection.
- Pixel-level prediction provides accurate segmentation.
- The architecture is suitable for real-time smart-city road-scene analysis.

Q3. Bidirectional RNN and Encoder–Decoder

A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

Answer:

A **Bidirectional RNN Encoder–Decoder** architecture can be used for multilingual customer-query translation while preserving contextual information.

Architecture:

Input Sentence → Embedding → Bidirectional RNN Encoder → Context Representation → RNN/LSTM Decoder → Attention → Target Sentence

Working:

1. Convert the input sentence into tokens.
2. Generate word embeddings.
3. The **Bidirectional RNN** processes the sentence in both:
 - Forward direction
 - Backward direction
4. The forward RNN captures previous context.
5. The backward RNN captures future context.
6. Their hidden states are combined to form a richer context representation.
7. The decoder generates the translated sentence one word at a time.
8. An attention mechanism can focus on the most relevant source words during each decoding step.
9. The model is trained using sequence loss such as categorical cross-entropy.
10. Translation quality can be measured using **BLEU score, accuracy, and human evaluation**.

Example:

English Query → BiRNN Encoder → Context Vector → Decoder → Hindi/Tamil/French Translation

Why Bidirectional RNN?

A traditional RNN processes information mainly in one direction. Bidirectional processing considers both past and future context, which is useful when word meaning depends on surrounding words. This helps the chatbot preserve contextual meaning during multilingual translation.

Q4. BPTT and LSTM

A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

Answer:

An **LSTM-based time-series prediction system** can be developed to forecast stock prices from historical market data. LSTM is preferred over a traditional RNN because it can preserve important long-term information and reduce vanishing-gradient problems.

Proposed Architecture:

Historical Stock Data → Preprocessing → Time-Series Sequences → LSTM Layer → Dropout → Dense Layer → Predicted Stock Price

Input Features:

- Open price
- High price
- Low price
- Closing price
- Trading volume
- Other technical indicators if required

Working:

1. Collect historical stock-market data.
2. Handle missing values and normalize numerical features.
3. Convert the data into fixed-length sequences.
4. Feed sequences into the LSTM network.
5. LSTM uses its **input, forget, and output gates** to control information flow.
6. The final hidden representation is passed to a Dense layer.
7. The network predicts the future stock price.
8. Calculate prediction error using **MSE or MAE**.
9. Compare predicted and actual prices using graphs.
10. Validate performance on unseen historical data.

Role of BPTT

Backpropagation Through Time (BPTT) unfolds the LSTM across time steps and calculates the error gradient through the sequence. The gradients are propagated backward through the unfolded network, and the weights are updated using an optimizer such as Adam.
Time t_1 → LSTM → t_2 → LSTM → t_3 → LSTM → Prediction → Loss → BPTT → Weight Update

Why LSTM over Traditional RNN?

- Handles long-term dependencies better.
- Reduces vanishing-gradient problems.
- Selectively remembers important information.
- Provides better temporal modelling than a basic RNN.

Q5. Integrated Deep Learning Solution

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

Answer:

An integrated video-captioning system can combine Transfer Learning for extracting visual information from video frames with an LSTM Encoder–Decoder for generating natural-language captions.

Proposed Architecture

Video → Frame Extraction → Pre-trained CNN → Visual Features → LSTM Encoder → Context Representation → LSTM Decoder → Word Prediction → Caption

Working:

1. Video Input:
Accept an uploaded video.
2. Frame Extraction:
Divide the video into meaningful frames at regular intervals.
3. Transfer Learning:
Use a pre-trained CNN such as ResNet50/EfficientNet to extract visual features from each frame.
4. Feature Extraction:
The CNN converts each frame into a compact feature vector.
5. LSTM Encoder:
The sequence of frame features is given to an LSTM encoder to learn temporal information.
6. Context Representation:
The encoder summarizes the visual sequence into a context representation.
7. LSTM Decoder:
The decoder receives the context and generates the caption word by word.
8. Word Prediction:
At each time step, the decoder predicts the next word using Softmax.
9. Caption Generation:
The process continues until an <END> token is generated.

Example:

Video of a person riding a bicycle

→ Generated Caption:

“A person is riding a bicycle on the road.”

Role of Each Component

Component	Role
Transfer Learning CNN	Extracts meaningful visual features
Frame Extraction	Converts video into a sequence of images
LSTM Encoder	Learns temporal relationships between frames
LSTM Decoder	Generates caption sequentially

Component	Role
Softmax	Predicts the next word
Attention mechanism	Focuses on important frames/features

Evaluation

The system can be evaluated using:

- BLEU score
- ROUGE score
- METEOR
- CIDEr
- Human evaluation
- Caption generation accuracy

Advantages:

- Reduces the requirement for training a CNN from scratch.
- Captures temporal relationships between video frames.
- Generates meaningful natural-language descriptions.
- Improves accessibility for users who cannot directly access video content.
- Can be extended with attention mechanisms and multilingual caption generation.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis